

Stereoselectivity in cyclic molecules

32

Connections

➡ Building on

- Stereochemistry [ch14](#)
- Conformational analysis [ch16](#)
- Saturated heterocycles and stereoelectronics [ch31](#)

Arriving at

- Stereoselectivity in cyclic systems is easy to understand
- Flattened four- and five-membered rings are attacked *anti* to large substituents
- Flattened six-membered rings are attacked from an axial direction
- Bicyclic structures are attacked on the outside face
- Tethering together nucleophile and electrophile forces one stereochemical outcome
- Hydrogen bonding can reverse the normal stereochemical outcome of a reaction

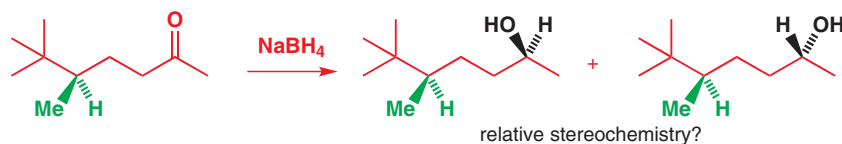
➡ Looking forward to

- Diastereoselectivity [ch33](#)
- Pericyclic reactions [ch34](#) & [ch35](#)
- Asymmetric synthesis [ch41](#)

Introduction

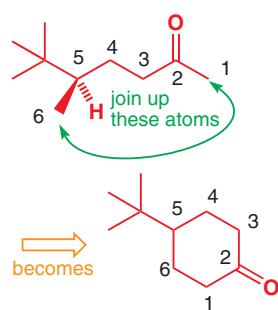
In the last chapter we looked at how the NMR spectra of cyclic molecules tell us a lot about their shape—both their conformation and their configuration. We are now going to go beyond simply studying stereochemistry and start to explain how to control stereochemistry. We have already, in Chapter 27, spent some time looking at controlling one aspect of stereochemistry—double bond geometry. But stereochemistry is about much more than this, and in this chapter and the next we will explain how to make single diastereoisomers and single enantiomers.

We start with stereochemistry in rings. Not only is stereochemistry easier to understand in cyclic compounds, it is also *better behaved* in cyclic compounds. Suppose you were to reduce this ketone to one of the corresponding alcohols.



To achieve a stereoselective reaction at the new stereogenic centre (shown in black) the green stereogenic centre would somehow have to influence the direction of attack of the nucleophile on the C=O group. Separated from it by three bonds, in a molecule with a high degree of flexibility, makes this a very tall order. A more or less 50:50 mixture of the two diastereoisomers would be expected.

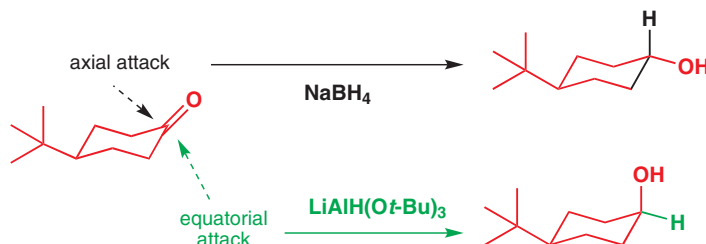
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.



■ If your memory of Chapter 16's discussion of the effect of substituents on the conformation of six-membered rings is dim, you should refresh it now. Conformational analysis underpins much of this chapter and you need to be fully familiar with its concepts and terminology.

➡ Stereoselective reactions of acyclic compounds are dealt with in Chapter 33.

However, if we join up the molecule into a ring, as shown in the margin, things are suddenly quite different. (This is not, of course, a chemical reaction—just a thought process!) The cyclic ketone has a fixed conformation controlled by the determination of the *tert*-butyl group to be equatorial. The two faces of the carbonyl group are therefore clearly quite different, and in fact by careful choice of reducing agent it is possible to attack either at will, giving almost exclusively either the axial or the equatorial alcohol. As we will explain shortly (p. 828) large reagents prefer to approach equatorially while small reagents prefer to approach axially, putting the new OH group into an equatorial position. These are *stereoselective* reactions and, because the two different outcomes are diastereoisomers, we can call them *diastereoselective*.

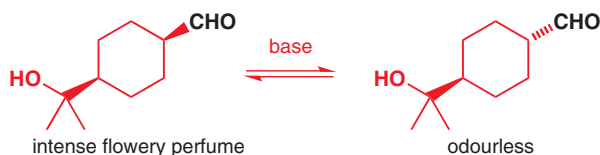


The key to the difference between these two compounds is in their conformations. The six-membered ring of the cyclic ketone has one conformation and the two approaches to the faces of the ketone are very different. In the open-chain compound rotation about all the C–C bonds is possible and very many conformations will be populated. In any one conformation, attack on one face of the ketone or the other may happen to be preferred, but summed over all of them the average selectivity will be close to 1:1. There is all the difference in the world between cyclic and open-chain compounds when it comes to stereoselective reactions.

In this chapter we shall look at reactions happening to cyclic compounds, reactions with cyclic intermediates, and reactions with cyclic transition states. We shall investigate what happens to stereochemistry when two (or even more) rings are joined together at a bond or at an atom. You have already looked in detail at reactions which close rings (Chapter 31, p. 805), and many of the *reactions* in this chapter you will have met earlier in the book. Our task is to reveal new features and subtleties, and to show you how to use these reactions to control stereochemistry.

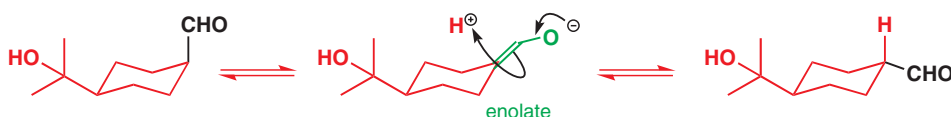
Stereochemical control in six-membered rings

As you saw in Chapter 16, cyclohexanes benefit from very well defined conformational preferences. Substituents are orientated either axially or equatorially, and usually prefer the equatorial orientation, especially when they are large. The strong preference for substituents to adopt the equatorial position means that when diastereoisomeric cyclohexanes equilibrate by processes such as enolization they may give high selectivity for the all-equatorial compound. For example, this fine perfumery material is made worthless by enolization.



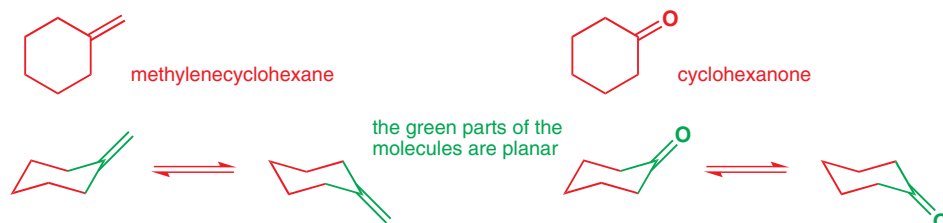
■ Although a disadvantage here, in other cases equilibration to the more stable all-equatorial conformation can be a useful source of stereochemical control. For an example, see p. 829.

The reason the equilibrium favours the worthless *trans* isomer (it forms 92% of the equilibrium mixture) is that the two substituents are both in the more stable equatorial positions.



Six-membered rings containing one sp^2 -hybridized carbon atom: cyclohexanone

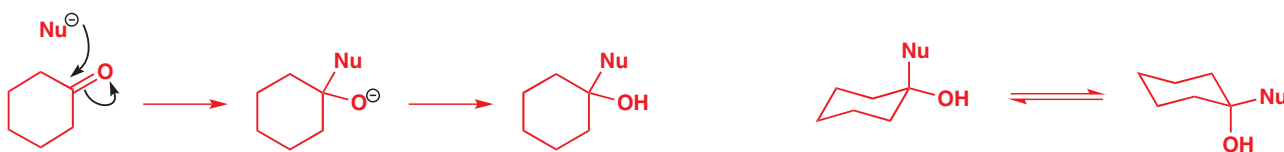
If we're interested in the reactions of six-membered rings, then we are going to have to consider what happens to their conformation when they contain reactive functional groups such as carbonyl groups and alkenes—in other words, the effect of introducing sp^2 C atoms into the ring. For just one sp^2 carbon atom the simple answer is that nothing changes—the conformation is not significantly altered by the presence of just one sp^2 centre in a ring. The conformations of methylenecyclohexene and cyclohexanone are shown below.



Six-membered rings with *more than one* sp^2 C atom do lose their chair conformation—they become flattened to some degree when there are one or more double bonds included in the ring and we shall come on to those in the next section.

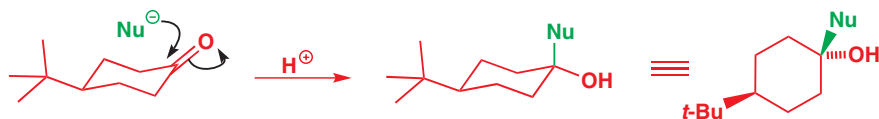
Axial or equatorial attack is possible on a cyclohexanone

So, what happens when a cyclohexanone is attacked by a nucleophile? For cyclohexanone itself, the reaction below gives a product which can adopt either of the two conformations shown, with Nu axial or equatorial, depending on the relative size of Nu and OH. This reaction does not tell us much about the attack on the C=O group itself—we can't tell, for example, whether Nu^- attacked the axial or the equatorial face of the C=O group.

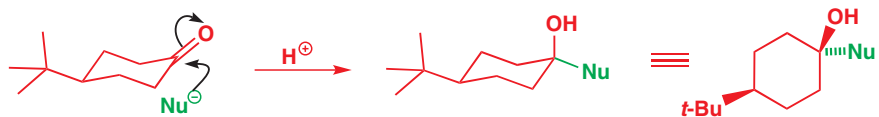


Now think of a nucleophile attacking 4-*t*-butylcyclohexanone. Since the *t*-butyl group locks the ring (*t*-Bu can never be axial), whether Nu is axial or equatorial will depend only on which face of the C=O group it attacks. Attack on the same face as the *t*-butyl group leaves the nucleophile axial and the hydroxyl group equatorial; attack on the opposite face leaves the nucleophile equatorial and the hydroxyl group axial. The nucleophile is said to attack either in an axial or equatorial manner, depending on where it ends up. It's easier to see this in a diagram.

axial attack of the nucleophile



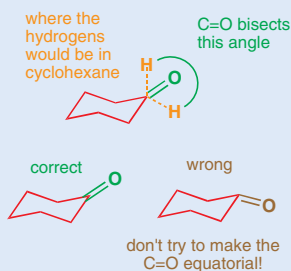
equatorial attack of the nucleophile



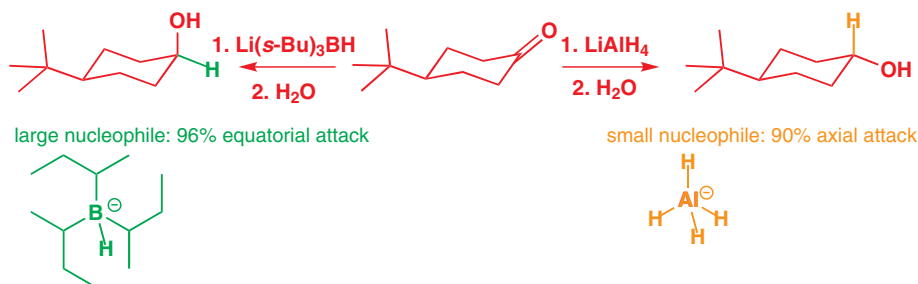
Now for an observation—we'll try and explain it shortly. In general, large nucleophiles attack equatorially and small nucleophiles attack axially. For example, reduction of 4-*t*-butylcyclohexanone with lithium aluminium hydride in Et_2O gives 90% of the *trans* alcohol: 90%

Drawing cyclohexanones

Make sure you point the ketone in the right direction! It should bisect the angle there would be between the axial and equatorial substituents, if the carbon atom were tetrahedral. It's always best to put the carbonyl group at one of the 'end' carbons of the ring: it's much harder to get it right if you join it to one of the middle ones.



of the hydride has added axially. AlH_4^- is quite small as nucleophiles go: to make more of the *cis* alcohol we need a larger nucleophile—lithium tri-*sec*-butylborohydride, for example, sold under the name of L-selectride®. This is so large that it attacks only equatorially, yielding typically 95% of the *cis* alcohol.



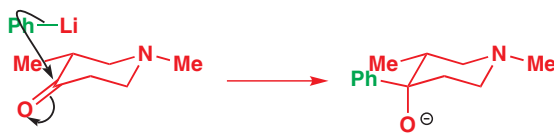
Carbon-centred nucleophiles follow the same trend—the table shows that, as size increases from the slender ethynyl anion through primary and secondary organometallics to *t*-BuMgBr, the axial selectivity drops off correspondingly. PhLi behaves as though it were quite small because it is flat.

Nucleophile	% of product resulting from	
	Axial attack	Equatorial attack
$\text{HC}\equiv\text{CLi}$	88	12
MeLi	35	65
PhLi	42	58
MeMgBr	41	59
EtMgBr	29	71
<i>i</i> -PrMgBr	18	82
<i>t</i> -BuMgBr	0	100

Now the difficult part—why? This is a question to which the answer really is not known for certain. It's certainly true that the direction of approach for axial attack is more hindered than for equatorial attack, and this is certainly the reason large nucleophiles prefer to attack equatorially.

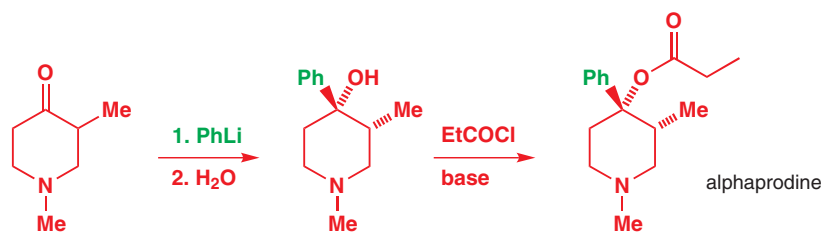
But if this is the case, why do small ones actually *prefer* to attack axially? There must be another factor that favours axial attack for those nucleophiles small enough to avoid the bad interactions with the other axial hydrogens. At the transition state, the forming O^- oxygen substituent is moving in either an axial or an equatorial direction. Just as the axial substituent is less favourable than an equatorial one, so is the transition state leading there, and the route leading to the equatorial hydroxyl group is favoured.

When chemists made the drug alphaprodine using the reaction shown below, they found that the combination of the equatorial preference of a methyl group adjacent to $\text{C}=\text{O}$ and an equatorial preference for attack on the $\text{C}=\text{O}$ group were enough to favour the formation of one diastereoisomer. Here is the reaction, with the starting material and product represented as a conformational diagram.

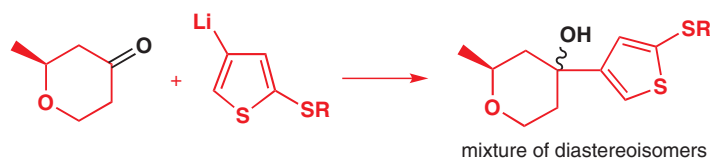


We can also represent the reaction in configurational terms. This is less good for explaining the stereoselectivity, but you should always be prepared to turn conformational diagrams into standard configurational ones.

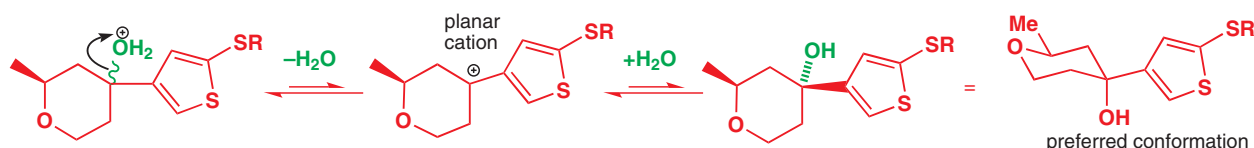
Other reasons have been proposed for this selectivity, but they are beyond the scope of this book.



In the next reaction, stereoselectivity is not so good. Zeneca (now AstraZeneca) announced the manufacture of a drug by the addition of a lithiated thiophene to another heterocyclic ketone, which initially gave a mixture of diastereoisomers.



Such a mixture is no good for manufacture of a pure drug, but the compound can be equilibrated in dilute acid by repeated S_N1 formation of a tertiary cation and recapture by water so that the required product (which is more stable as it has both Me and the thiophene equatorial) dominates by 92:8 and can be purified by crystallization. The unwanted isomer can be recycled in the next batch.



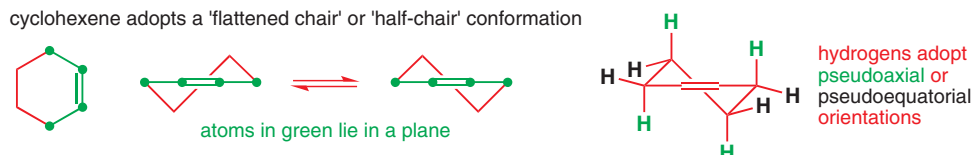
■ Compare this strategy with the equilibration of the perfumery compound via its enolate, described on p. 826.

In these reactions the molecule has a free choice whether to place a substituent in an axial or equatorial position and this is the only consideration because the starting materials in the reactions—ketones or carbocations—have six-membered rings that are already in the chair conformation even though they have one trigonal (sp^2) atom in the ring.

Six-membered rings with two or more sp^2 carbons: cyclohexenes

With more than two trigonal carbon atoms in the ring, a cyclohexene can no longer adopt a chair conformation. At least four of the atoms in the ring must now be in a plane, and the best way to represent this is in the diagrams shown below. The four atoms in the plane are nearest you, with the remaining two placed one above and one below that plane.

cyclohexene adopts a 'flattened chair' or 'half-chair' conformation



Cyclohexene itself flips rapidly between these two conformations, with a barrier of about 22 kJ mol^{-1} , about half that of cyclohexane. As in cyclohexane, hydrogen atoms on the saturated carbons of the cyclohexene structure adopt two types of positions, but as they are not quite orientated in the same way as in cyclohexane, we call the two orientations 'pseudoaxial' and 'pseudoequatorial'.

Only axial attack is possible with cyclohexenes

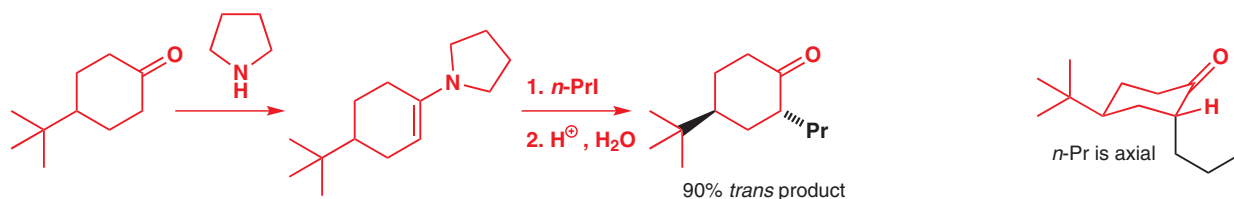
These conformations of six-membered rings with more than one trigonal carbon are quite plainly not chairs, and are much less stable than chairs. Anything which allows them to become a chair is likely to be highly favoured, and the stereoselectivity of a reaction is likely

to be driven by the need for the transition state and product to have a chair rather than a boat conformation. This can override the preference for substituents to go into equatorial positions. The choice of axial attack controls the stereoselectivity of reactions of cyclohexenes (and, as you will see, their epoxides), six-membered cyclic enolates, and six-membered cyclic enones.

● **The number of trigonal carbon atoms in the ring decides which factors control stereoselectivity**

- Six-membered rings with one trigonal (sp^2) carbon atom are already chairs and can undergo *axial or equatorial attack*.
- Six-membered rings with two or more trigonal carbon atoms are not chairs and undergo *axial attack* in order to form chairs rather than boats. The final product may end up with axial or equatorial substitution, but this is not a consideration in the reaction itself.

Interactive mechanism for axial alkylation of cyclohexanone enamine

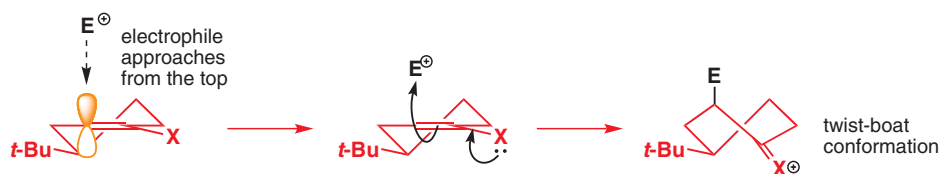


Enolate equivalents were discussed in Chapter 25.

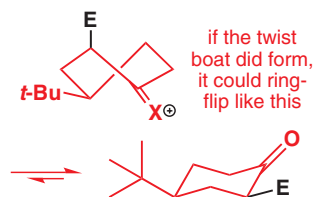


To get at the reason for this result we need to look at the conformation of the enamine intermediate. At this point we shall generalize a bit more and write a structure that represents any enol derivative where X may be OH, O[−], OSiMe₃, NR₂, and so on. The double bond ($2 \times sp^2$ centres) in the ring means the conformation is a partially flattened chair, as described above. We place the *t*-butyl group in an equatorial position because, as with cyclohexanes, it is so bulky it cannot go axial. This means that there is only one conformation to consider—the one shown in the margin.

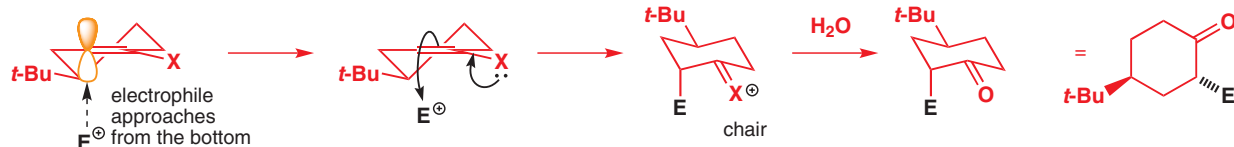
Now, the electrophile must attack the π system of this conformation, and to do so it has to attack from more or less directly above or below because only then can it interact with one of the lobes of the p orbital at the enol position shown in orange. The need to interact with the π system is the reason cyclohexenes and related compounds react in an axial direction. The top of the molecule looks to be more open to attack so we shall try that approach first.



As the electrophile forms a bond to the trigonal carbon atom, that atom must become tetrahedral and it does so by forming a vertical bond upwards. The result is shown in the diagram—the ring turns into a twist-boat conformation. Now, of course, after the reaction is over, the ring can flip into a chair conformation and the new substituent will then be equatorial, but that information is not present in the transition state for the reaction. We could say that, at the time of reaction, the molecule doesn't 'know' it can later be better off and get the substituent equatorial: all it sees is the formation of an unstable twist boat with a high-energy transition state leading to it.



Attack from the apparently more hindered bottom face makes the trigonal carbon atom turn tetrahedral in the opposite sense by forming a vertical bond to the electrophile *downwards*. The ring goes directly to a chair form with the electrophile in the axial position.

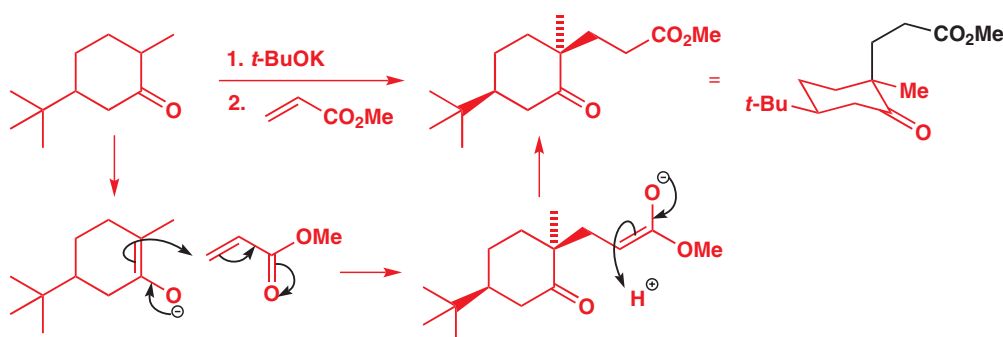


When the carbonyl group is restored by hydrolysis (if necessary—with an enolate X is already O) the ring need not flip: it's already a chair with the t -butyl equatorial, and the new substituent is axial on the chair. This is the observed product of the reaction.

It's important that you understand what is going on here. The reagent *has* to attack from an axial direction to interact with the p orbital. If it attacks from above, the new substituent is axial on an unstable twist boat. If it attacks from below, the new substituent is axial on a chair—granted, this is not as good as equatorial on a chair, but that's not an option—it has to be axial on something, and a chair is better than a twist boat. So this is the product that forms. It's just hard luck for the substituent that it can't know that *if it did* weather it out on the twist boat it could later get equatorial—it plumps for life on the chair and so has to be content with ending up axial.

Here is an example with an unsaturated carbonyl compound as an electrophile: the reaction is a Michael addition. The ketone here is slightly different—it has the t -butyl group in the 3- rather than the 4-position, and the reacting centre becomes quaternary during the Michael reaction. But the result is still axial attack.

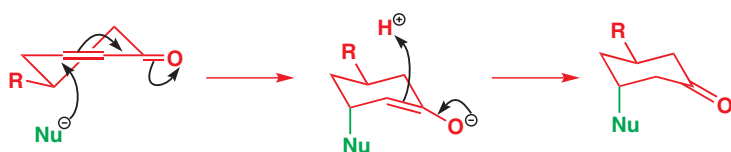
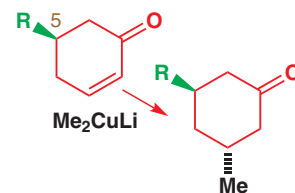
➔ Such reactions were discussed in Chapter 25.



This result is more impressive because the large electrophile ends up on the *same* side of the ring as the t -butyl group, so the stereoselectivity cannot be based on any simple idea of reaction on the less hindered side of the ring. It is genuine axial attack, as the conformational diagram of the product confirms.

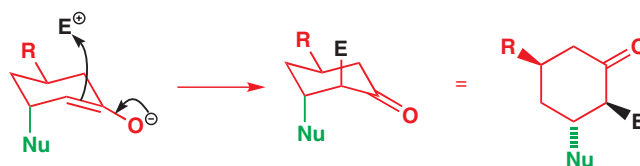
Cyclohexenones are even flatter than cyclohexenes, but it is convenient to draw them in a similar conformation. Conjugate addition to the substituted cyclohexenone in the margin gives the *trans* product.

This is also axial addition to form a chair directly (rather than a twist boat) with the nucleophile approaching from the bottom. We must draw the ring as a flattened chair.

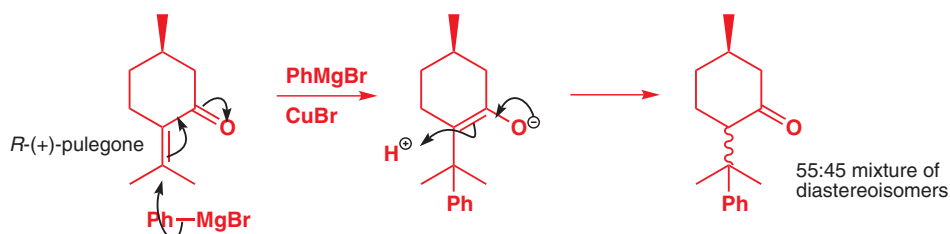


The 5-alkyl cyclohexenone that we have chosen as our example gives the best results. The mechanism suggests that the enolate intermediate is protonated on the top face (axial addition again), although we cannot tell this because the product has no stereogenic centre there. But, if we carry out a tandem reaction with the enolate trapped by a different electrophile, it becomes clear that the product is again that of axial attack.

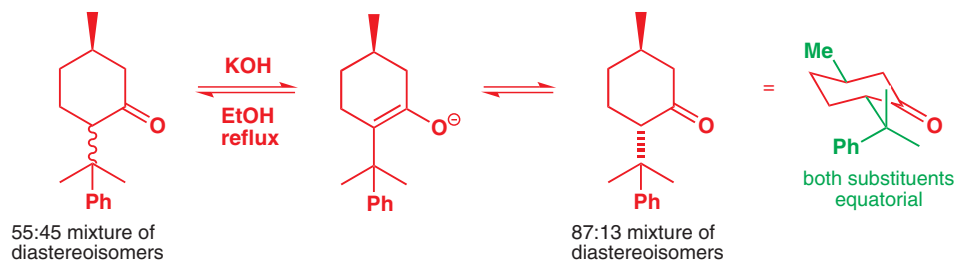
■ Beware: you also get the same answer but for the wrong reason by saying that the nucleophile approaches from the less hindered side.



We shall end this section on conformational control in six-membered rings with the preparation of a useful chiral molecule, 8-phenylmenthol, from the natural product (*R*)-(+)-pulegone. The first step is a copper-promoted conjugate addition to an exocyclic alkene. A new stereogenic centre is formed by protonation of the enolate intermediate but with virtually no stereoselectivity.

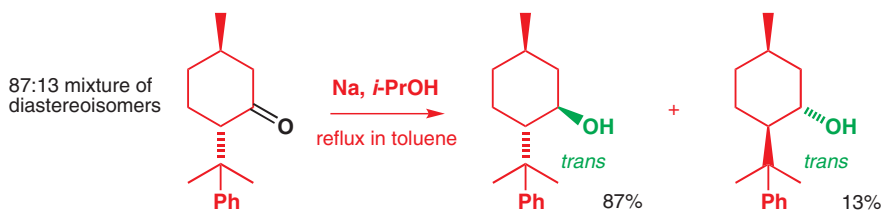


Now thermodynamic control can be brought into play. The position next to the ketone can be epimerized via the enolate to give the more stable isomer with both substituents equatorial. This improves the ratio of diastereoisomers from 55:45 to 87:13.



Now the ketone can be reduced with a small reagent (see p. 826)—Na in *i*-PrOH works well—to put the hydroxyl group equatorial. This means that all the product has OH *trans* to the large group next to the ketone, although it is still an 87:13 mixture of diastereoisomers with respect to the relative configuration at the centre bearing Me.

■ Na in *i*-PrOH is a single electron Birch-type reduction (see Chapter 23). You can't get much smaller than an electron!



■ The product, 8-phenylmenthol, can be used as a chiral auxiliary. See Chapter 41, p. 1113.

These alcohols can be separated (they are, of course, diastereoisomers and not enantiomers) and the major, all-equatorial, one is the useful one. This is an impressive example of conformational control by thermodynamic and by kinetic means originating only from a distant methyl group in a six-membered ring.

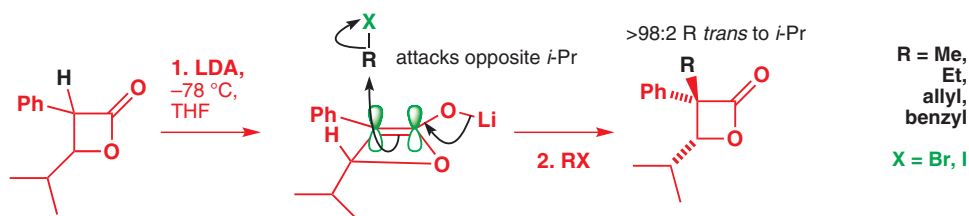
Reactions on small rings

The conformational principles which apply to rings other than six-membered ones are rather more sketchy because only six-membered rings adopt well-defined chair (or, for cyclohexenes, half-chair) conformations. But we can still give you some general guidelines and principles, and illustrate them with some important examples. We will look in detail at four- and five-membered rings.

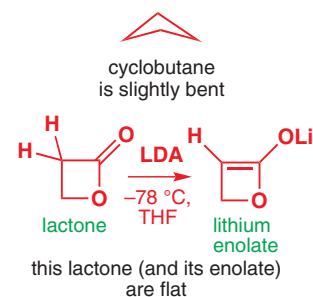
➡ We looked at the effect of ring size on NMR spectra and on rates of ring closure in Chapter 31, see pp. 814 and 805.

Four-membered rings can be flat

Saturated four-membered rings have a slightly bent conformation but four-membered lactones are flat. The enolates of these lactones can be made in the usual way with LDA at -78°C and are stable at that temperature, and they react with electrophiles just as you saw in Chapter 25. If the β -lactone has a substituent already then there may be a choice as to which face of the enolate is attacked by an electrophile. In the example below, simple alkylation with a variety of alkyl halides gives essentially only one diastereoisomer of the product.



The enolate, as we have seen, is planar, and the phenyl group is in the plane (which is why it doesn't matter which of the two possible diastereoisomers of the starting material is used). The isopropyl group is the only thing out of the plane. The electrophile simply adds to the face of the enolate not blocked by the isopropyl group. This is a very simple case of a diastereoselective reaction.

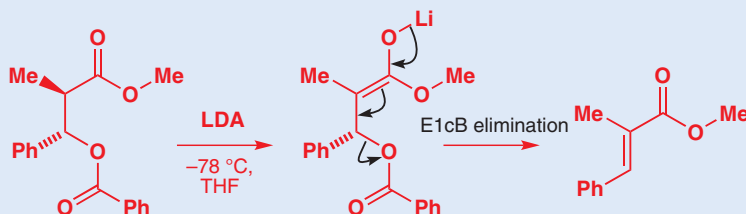


Diastereoselective reactions of racemic compounds

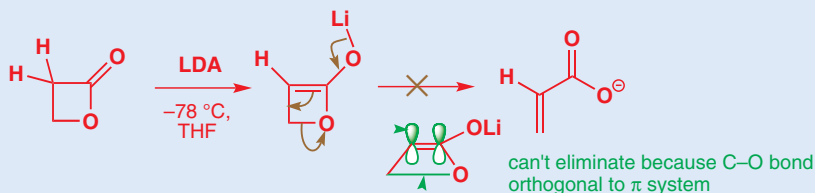
The stereoselectivity we are discussing in this chapter is *diastereoselectivity*; we are not concerned with enantiomers and all of our discussions are equally valid whether the starting materials are racemic or enantiomerically pure. The product here, as in many other examples in the chapter, is racemic so we could write ($\pm$) underneath the structure. In this particular reaction, the starting material can be either of two diastereoisomers, but one of its chiral centres is lost on formation of the enolate.

Lactone enolates

This lithium enolate works well even though it might be expected to be unstable because of a simple elimination reaction. In general, it is not possible to make open-chain lithium enolates with β oxygen substituents like this because they do undergo elimination.



But, in the four-membered ring, the p orbitals of the enolate and the C–O single bond are orthogonal (see diagram below) so that no interaction between them, and no elimination, can occur. In the terminology of Baldwin's rules (Chapter 31, p. 810) it would be a disfavoured 4-*endo-trig* reaction.

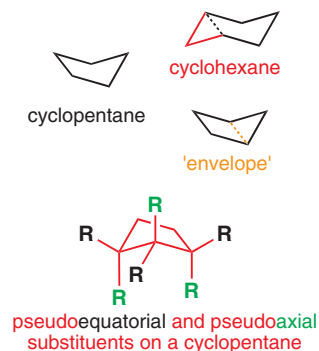


Reduction of substituted four-membered ring ketones is usually reasonably stereoselective. If the substituent is in the 3-position and small reagents like NaBH_4 are used, the *cis* isomer is favoured. Like saturated four-membered rings, cyclobutanones are slightly puckered to reduce eclipsing interactions between hydrogen atoms on adjacent carbon atoms, but attack of the reducing agent still occurs from the direction away from the other substituent to give the *cis* product.



Five-membered ketones are flexible

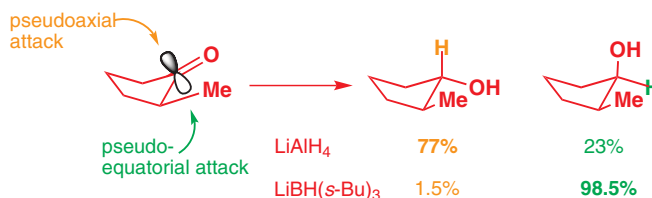
➡ We discussed this conformation and the consequences for NMR of its flexibility in Chapter 31, p. 817.



➡ We discussed the direction of attack on cyclohexanones on p. 826.

A saturated five-membered ring has a conformation often called an 'envelope'. It looks a bit like an opened envelope with one atom at the point of the flap. The arrangement closely matches what you get if you cut one atom out of a cyclohexane ring. At any one moment, one of the carbon atoms is at the point of the envelope but rapid ring flipping equilibrates all these conformers so that all five atoms are, on average, the same.

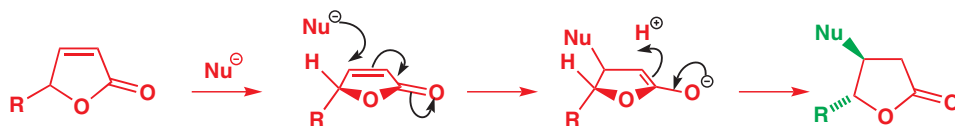
Substituted cyclopentanes can have substituents in pseudoaxial or pseudoequatorial positions (in other words, they are somewhat like the axial and equatorial positions in a cyclohexane), but rapid equilibration means that overall we have a very flexible and labile system. As a result, reduction of 2-substituted cyclopentanones may not be very stereoselective. What selectivity there is (about 3:1) in the reduction of 2-methylcyclopentanone with LiAlH_4 favours pseudoaxial attack in the conformation drawn, as is reasonable for a small nucleophile.



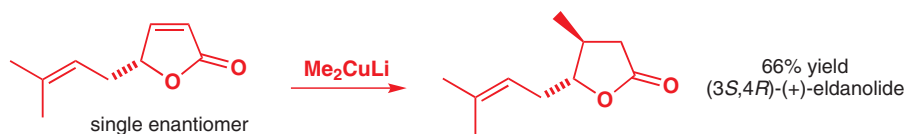
The use of a much more bulky reducing agent such as LiBH(s-Bu)_3 dramatically reverses and increases the stereoselectivity. Essentially only the *cis* compound is formed.

Regard five-membered rings with two or more sp^3 carbons as flat

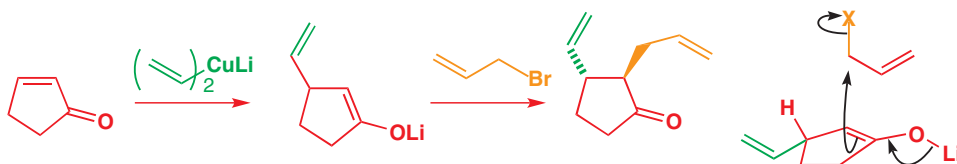
When there are two or three trigonal carbons in the ring, the ring is flatter and reactions such as enolate alkylation and conjugate addition give excellent stereoselectivity even with a simple cyclopentane ring. Unsaturated five-membered lactones (known as 'butenolides') give a very clear illustration of stereochemically controlled conjugate addition. There is only one possible stereogenic centre and the ring is almost planar so we expect nucleophilic attack to occur from the less hindered face. Cuprates are good nucleophiles for this reaction and here Me_2CuLi adds to the unsaturated lactone.



With a single enantiomer of the starting material below, the product is the single enantiomer of an insect pheromone.



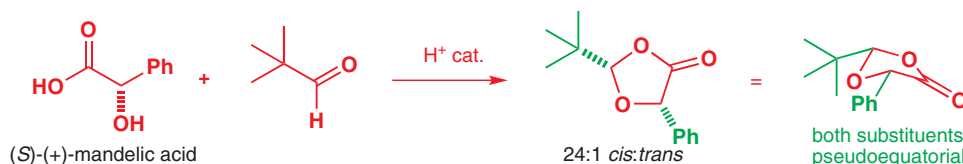
It is not even necessary to have a stereogenic centre in an unsaturated ring if we want to create stereochemistry. A tandem conjugate addition and alkylation creates two new stereogenic centres in one operation. The conjugate addition of a lithium cuprate makes a lithium enolate, which will react in turn with an alkyl halide. The product is usually *trans*.



■ This would be a good point at which to remind you of what we stressed in Chapter 14. If all the starting materials and reagents are achiral or racemic the products must be too. That has been the case in many of the reactions so far in this chapter: we haven't put in ($\pm$) under every compound but we could have done. But here we do have a single enantiomer of starting material so we get a single enantiomer of product. Diastereoselectivity is the same whether the starting material is enantiomerically pure or racemic.

The key step is the alkylation of the enolate intermediate. Enolates in five-membered rings are almost flat and the incoming orange allyl bromide prefers the less hindered face away from the recently added green vinyl group.

Our main example of enolate reactions in five-membered rings is one of some general importance. It illustrates how stereochemical information can be transmitted across a ring even though the original source of that information may be lost during the reaction. That may sound mysterious, but all will become clear. The first reaction is to make a five-membered cyclic acetal from an optically active hydroxy-acid. Our example shows (*S*)-(+)-mandelic acid reacting with *t*-BuCHO.



■ The conjugate addition forms a lithium enolate regiospecifically, and that was why you met this sequence in Chapter 25. We showed you a dramatic use of the stereoselectivity there as well, in a synthesis of a prostaglandin (p. 604).

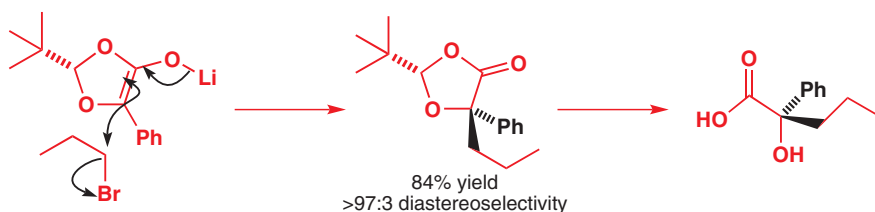
■ Check that you can write the mechanisms for acetal formation (Chapter 11). Acetal formation is under thermodynamic control so the product produced is the more stable.

Acetal formation involves nucleophilic attack of the OH group on the aldehyde so there is no change at the stereogenic centre. The stereochemistry of the new (acetal) centre may surprise you—why should the *cis* isomer be so favoured? This is a conformational effect as both substituents can occupy pseudoequatorial positions.

Now, if we make the lithium enolate with LDA, the original stereogenic centre is destroyed as that carbon becomes trigonal and planar. The only stereogenic centre left is the newly introduced one at the acetal position.

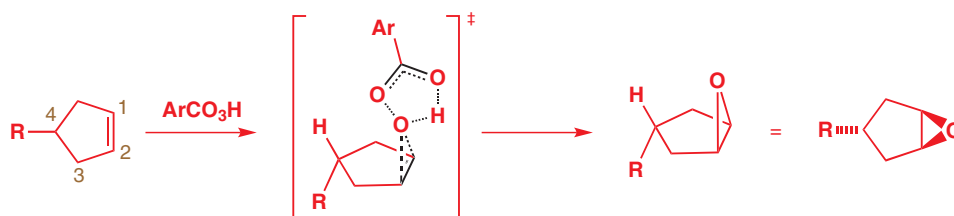


The ring is now essentially flat, owing to the C=C bond within it, and reaction of the enolate with an electrophile is again a simple matter of addition to the face of the enolate opposite to the *t*-butyl group.



If the acetal is now hydrolysed, the new stereogenic centre is revealed as an alkylated version of the starting material. It may appear that the alkylation has happened stereospecifically with retention, but what has really happened is that the new stereogenic centre in the acetal intermediate has relayed the stereochemical information through the reaction.

Five-membered rings also allow us to explore electrophilic attack on alkenes. A simple 4-substituted cyclopentene has two different faces—one on the same side as the substituent and one on the opposite side. Epoxidation with a peroxy-acid occurs preferentially on the less hindered face.

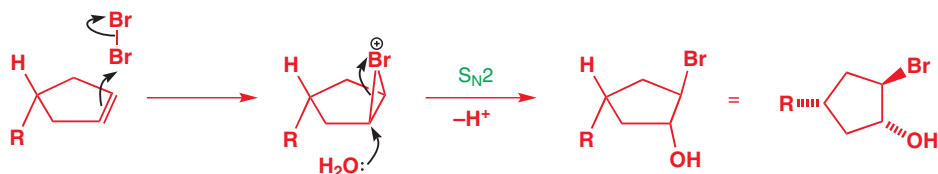


➡ The mechanism of RCO_3H epoxidation was discussed on p. 430.

■ Note that this reaction is diastereoselective—but neither starting material nor products are chiral. Diastereoselectivity need have nothing to do with chirality!

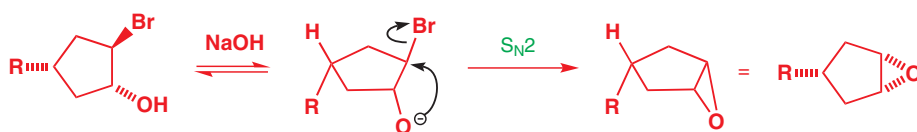
In the transition state (marked ‡) the peroxyacid prefers to be well away from R, even if R is only a methyl group (the selectivity is 76:24 with R=Me).

The opposite stereoselectivity can be achieved by bromination in water. The bromonium ion intermediate is formed stereoselectively on the less hindered side and the water is forced to attack stereospecifically in an S_N2 reaction from the more hindered side.



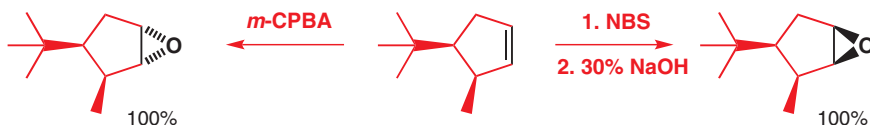
■ You will spot that this reaction is no longer bimolecular because the nucleophile and leaving group are part of the same molecule. We still call it S_N2 because the pathway of the mechanism is identical with a normal S_N2 reaction. Substitution reactions were discussed in detail in Chapter 15.

Treatment of the product with base (NaOH) gives an epoxide by another S_N2 reaction in which oxygen displaces bromide. This is again stereospecific and gives the epoxide on the same side as the R group.



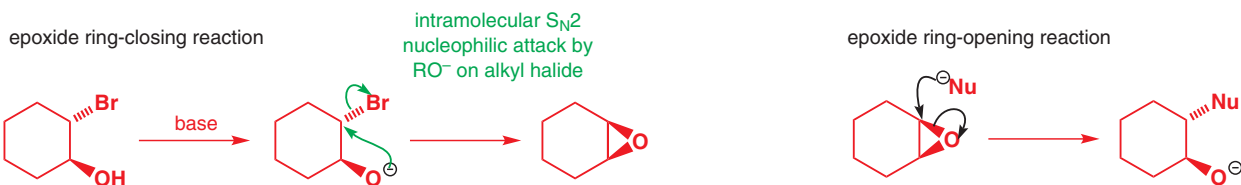
Two substituents on the *same side* of a five-membered ring combine to dictate approach from the other side by any reagent, and the two epoxides can be formed each with essentially 100% selectivity.

➡ *N*-Bromosuccinimide (NBS) acts as a source of electrophilic bromine: see Chapter 19, p. 441.



Regiochemical control in cyclohexene epoxides

The two reactions above illustrate two important ways of making an epoxide. We are now going to look in a little more detail at what happens when epoxides are opened—a reaction that is essentially the reverse of the epoxide-closing reaction you have just seen. Here are both reactions with the epoxide fused to a cyclohexane ring:

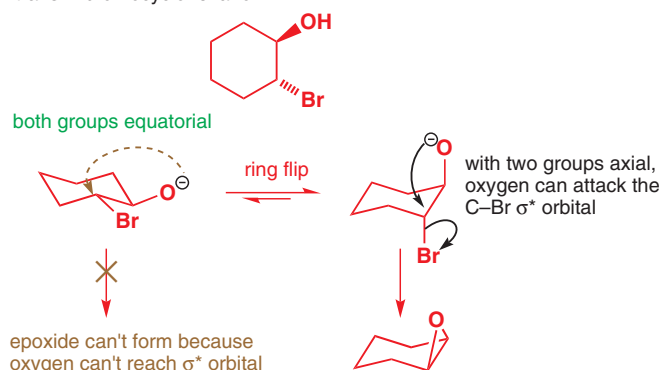


Epoxides can be formed from compounds containing an adjacent hydroxyl group and a leaving group by treatment with base. The epoxide formation is an intramolecular S_N2 reaction, and as with any S_N2 substitution, inter- or intramolecular, the incoming nucleophile must still attack into the σ^* orbital of the leaving group. And the only way that can happen, as you can see from the diagrams below, is (a) if the hydroxyl group and leaving group are *trans* to one another and (b) if the hydroxyl group and leaving group are both orientated axially. For the *trans* diastereoisomer, the groups can of course adopt either a diequatorial or a diaxial arrange-

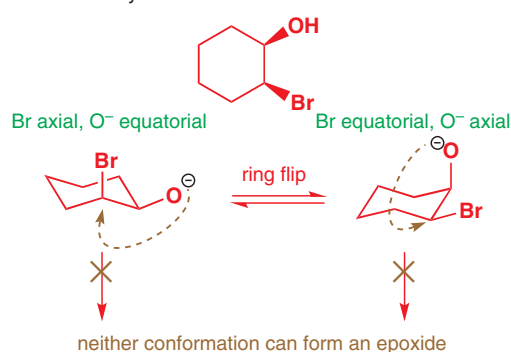
ment (the diequatorial arrangement is favoured, as you saw in Chapter 16) but only the diaxial can react. The *cis* diastereoisomer cannot form an epoxide.

■ In Chapter 36 you will meet the alternative **rearrangement reactions** that occur if you try to force *cis* substituted compounds like these to react.

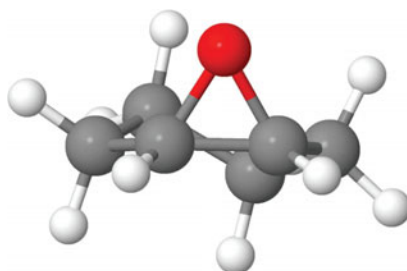
trans-2-bromocyclohexanol



cis-2-bromocyclohexanol

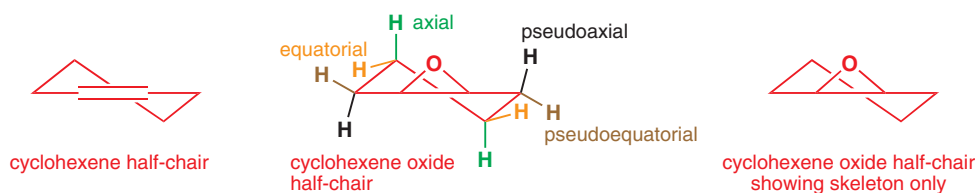


How should we draw this epoxide fused to a six-membered ring? It is impossible for the CO bonds of the product epoxide ring to adopt perfectly axial and equatorial positions. If you make a model of cyclohexene oxide (as we can call this epoxide) you will see that the ring is a slightly deformed chair—in fact it is like the half-chair conformation of cyclohexene, in which four of the carbon atoms are in the same plane (you met this on p. 829).

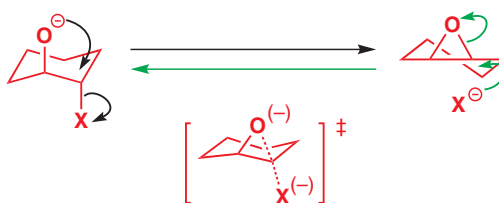


model of cyclohexene oxide

The usual way of drawing cyclohexene oxide is shown below: the distortion due to the three-membered ring changes the orientation of the axial and equatorial hydrogens next to the ring—they are **pseudoaxial** and **pseudoequatorial**. The hydrogens on the back of the ring (this part of the ring remains about the same as in the chair conformation) can be still considered as 'normal' axial and equatorial hydrogens.

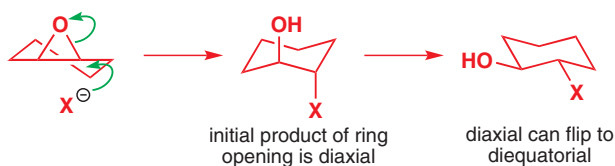


You saw above that the epoxide-forming reaction is essentially the reverse of the epoxide-opening reaction. If we took a snapshot of the transition state for either reaction, we would not be able to tell whether it was the RO^- that was attacking the C-X σ^* orbital to form the epoxide with X^- as a leaving group, or a nucleophile X^- attacking the C-O σ^* orbital of the epoxide to form a ring-opened alcohol. In other words, the transition state is the same for both reactions.



this transition state is the same for both formation and ring opening of the epoxide

Since ring closure is possible only when the starting material is diaxially substituted, this has to mean that ring opening is similarly possible only if the *product* is diaxial. This is a general principle: *ring opening of cyclohexene oxides always leads directly to diaxial products*. The diaxially substituted product may then subsequently flip to the diequatorial one, but it is always the one that is initially formed.



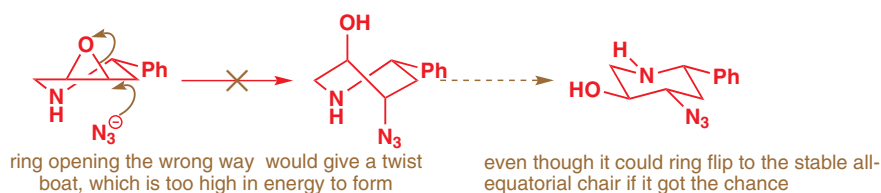
How do we know this to be true? If the ring bears a bulky substituent, ring flipping is impossible and the diaxial product has to stay diaxial. An example is nucleophilic attack of halide on the two epoxides shown below. The fact that the ring is a piperidine, rather than a cyclohexane, does not matter. The equatorial phenyl group fixes the conformation, and the regiochemistry of the epoxide opening with azide depends only on the relative stereochemistry of the starting material.



Points to note:

- The nucleophile must attack from the opposite side of the epoxide, allowing it to put electrons into the C–O σ^* orbital. This means that the nucleophile and hydroxyl group always end up *trans* in the product.
- The phenyl group locks the conformation of the epoxide. It stays equatorial, so we only have one epoxide conformation to consider in each case.
- In each case the epoxide opens only at the end that gives the diaxially substituted chair. Ring opening at the other end would still give a diaxially substituted product, but it is a diaxially substituted high-energy twist-boat conformation. The twist boat can, in fact, flip to give an all-equatorial product, but in a kinetically controlled

process such as this, it is the barrier to reaction that matters, not the stability of the final product.



● **Some general observations on stereo- and regioselectivity in six-membered rings:**

- Six-membered rings which are not already a chair (such as cyclohexenes and cyclohexene oxides) react in such a way that they immediately become a chair.
- They do so by reacting from an axial direction: this may also dictate the *regioselectivity* of the reaction.
- Six-membered rings which are chairs already (such as cyclohexanones) remain a chair, and react from either the axial or equatorial direction according to the size of the attacking reagent.

Stereoselectivity in bicyclic compounds

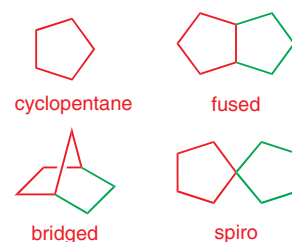
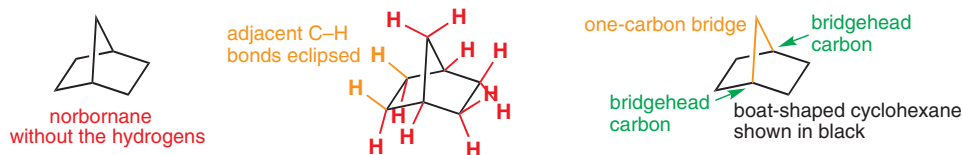
We have just looked at the way the reactivity of an epoxide gains additional subtleties when it is fused into a bicyclic structure with a six-membered ring. We're now going to look more generally at bicyclic compounds and their reactivity, and consider some features of their stereoselective reactions.

Bridged bicyclic rings

There are broadly three kinds of bicyclic compounds. If we imagine adding a second five-membered ring to one already there, we could do this in a bridged, fused, or *spiro* fashion, as you see in the margin. Bridged bicyclic compounds are just what the name implies—a bridge of atom(s) is thrown across from one side of the ring to the other. Fused bicyclic compounds have one *bond* common to both rings, while *spiro* compounds have one *atom* common to both rings.

You will notice that these three types of bicyclic compounds with five-membered rings have different numbers of atoms added to a 'parent' five-membered ring. The bridged compound has two extra atoms, the fused compound three, and the spiro compound four. These are marked in green with the original five-membered ring in red. We shall consider stereoselectivity in each of these types of bicyclic ring systems, starting with bridged structures.

The bridged ring shown in the margin is known as norbornane: it's a simple but very important skeleton on which many other structures are based, and it's worth spending a moment learning how to draw it convincingly. The instructions in the box overleaf tell you how! Another way of looking at norbornane is as a six-membered ring held in a boat conformation by a one-carbon bridge. The bridge has to be axial at both bridgehead positions (or it wouldn't be able to form a ring) so the cyclohexane has no choice but to be a boat.

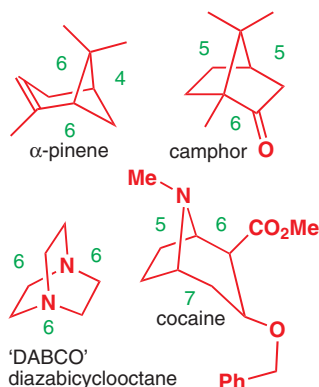
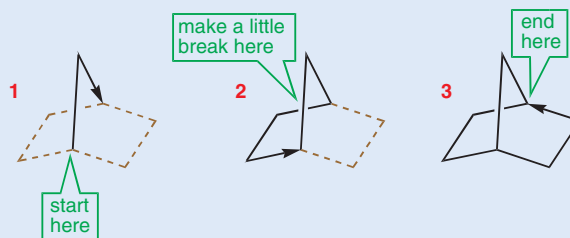


Naming bicyclic compounds

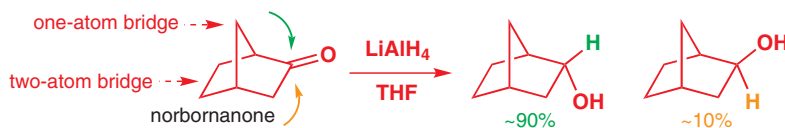
As usual we shall not spend too long on nomenclature, but you may hear norbornane structures referred to as 'bicyclo[2.2.1]heptanes'. The 'bicyclo' and 'heptane' parts are self explanatory. The numbers (always separated by dots) refer to the lengths of the bridges linking the two bridgehead carbons. The other two compounds in the margin above are thus bicyclo[3.3.0]octane and spiro[4.4]nonane.

How to draw norbornane structures

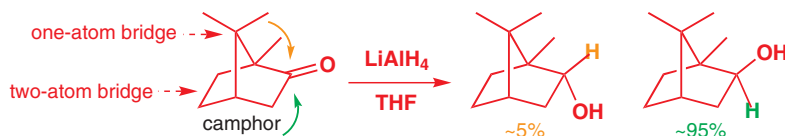
The easiest way to draw a convincing norbornane is to start with the bridge: draw a sort of skewed upwards chevron as shown in **1**. Then join the ends of the chevron with three bonds, as in **2**, making sure to break one of them as it passes behind the chevron, to give an impression of the three-dimensional shape of the molecule. Finally, link the second ring round to the right, **3**.



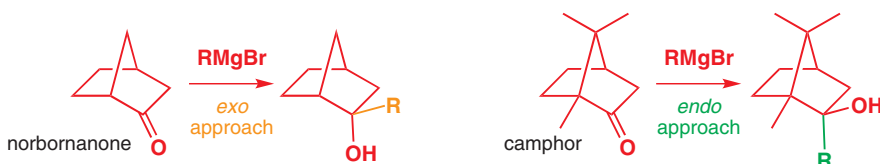
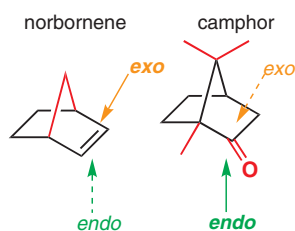
A selection of important bridged bicyclic compounds is shown in the margin. Bridged structures (sometimes called cage structures) are generally very rigid, spending most of their time in a single, well-defined conformation, and this rigidity is reflected in the stereochemistry of their reactions. For example, attack on norbornanone occurs predominantly from the side of the one-atom bridge (the green arrow) rather than the two-atom bridge (the orange arrow).



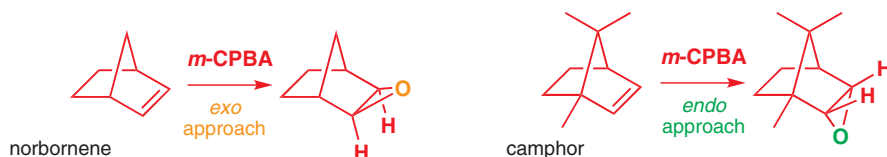
This selectivity is completely reversed in camphor because the one-atom bridge then carries two methyl groups. One of these must project over the line of approach of the hydride reducing agent.



The two methyl groups on the bridge of the camphor molecule are key features in stereoselective reactions—take them away and the result often changes dramatically. This bicyclic system, with and without methyl groups, has been so widely used to establish stereochemical principles that the two faces of, say, the ketone group in camphor, or the C=C double bond in norbornene (the alkene derivative of norbornane) have been given the names *endo* and *exo*. These refer to inside (*endo*) and outside (*exo*) the boat-shaped six-membered ring shown in black. In general, reactions of norbornane-type structures occur from the less hindered *exo* face, but the methyl groups of camphor reverse this selectivity to favour *endo* attack:



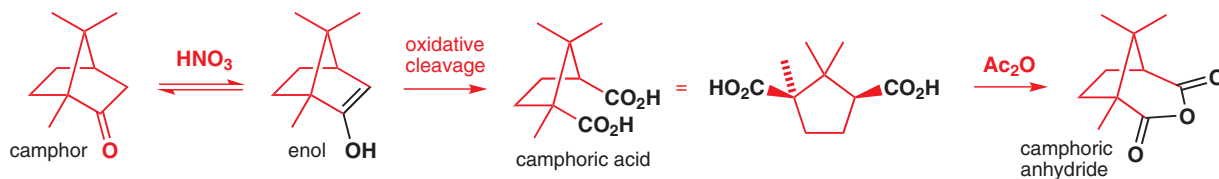
In a similar style, epoxidation of the two alkenes is totally stereoselective, occurring *exo* in norbornene and *endo* when methyl groups are present on the bridge. These stereoselectivities would be remarkable in a simple monocyclic compound, but in a rigid bridged bicyclic structure they are almost to be expected.



Reactions that break open bridged molecules can preserve stereochemistry

Some powerful oxidizing agents are able to cleave C–C bonds. Oxidation of camphor with concentrated nitric acid cleaves a C–C bond adjacent to the C=O group and produces a diacid known as camphoric acid. The usual reagent is nitric acid (HNO_3) and oxidation goes via camphor's enol.

➡ This is an unusual reaction; more common is cleavage of C=C bonds with ozone, as you saw in Chapter 19.

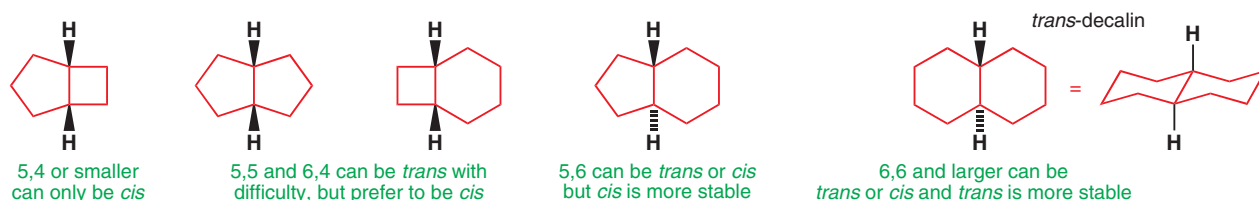


Because the bridge holds the molecule in a fixed conformation, the cleaved diacid has to have a specific stereochemistry. There is no change at the stereogenic centres, so the reaction must give retention of configuration. We can confidently write the structure of camphoric acid with *cis*-CO₂H groups, but any doubt is dispelled by the ability of camphoric acid to form a bridged bicyclic anhydride.

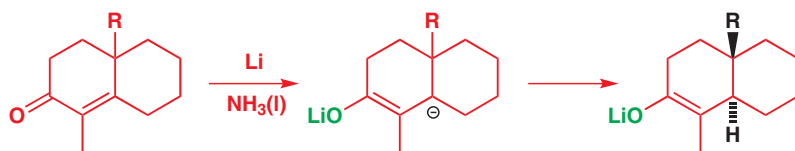
Fused bicyclic compounds

trans-Fused rings

The ring junction of a fused 6,5-membered ring system can have *cis* or *trans* stereochemistry, and so can any pair of larger rings. For smaller rings, *trans* 5,5- and 6,4-ring junctions can be made, with difficulty, but with smaller rings *trans* ring junctions are essentially impossible.

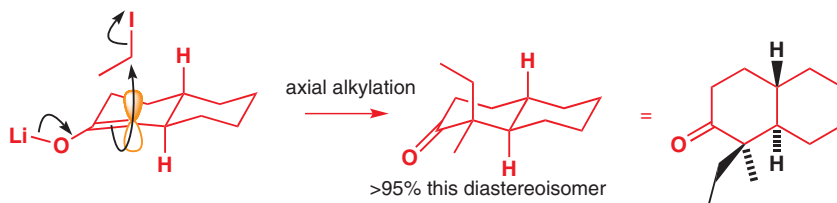


The *trans*-fused 6,6 systems—*trans*-decalins—have been very widely studied because they form an important part of the structure of steroids. Their conformation was discussed in Chapter 16: they prefer a *trans* ring junction as *trans*-decalins have all-chair structures with every bond staggered from every other bond. We can show this by giving a 6,6 system the choice: reducing this enone with lithium metal gives a lithium enolate (Chapter 25). Protonation of this anion with the solvent (liquid ammonia) gives a *trans* ring junction.

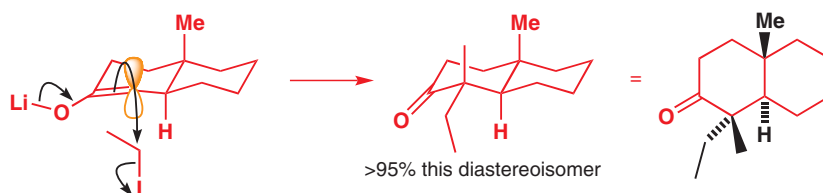


The lithium enolate remains and can be alkylated with an alkyl halide in the usual way. When there are hydrogen atoms at both ring junction positions, axial alkylation occurs just as you should now expect, and a new ketone with three stereogenic centres is formed with >95% stereoselectivity.

■ In this scheme, and the next, the methyl group attached at the yellow p orbital has been omitted for clarity.

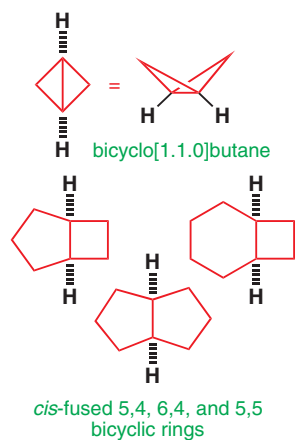


However, if there is anything else—even a methyl group—at the ring junction, so that axial approach would give a bad 1,3-diaxial interaction in the transition state, the usual stereoselectivity is overridden and the reaction switches to alkylation on the other face:



cis-Fused rings

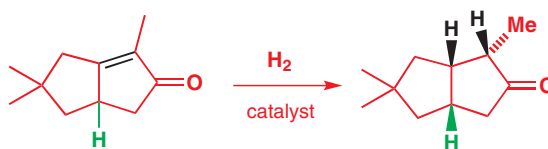
➡ See the box on p. 839 for an explanation of the name bicyclo[1.1.0]butane.



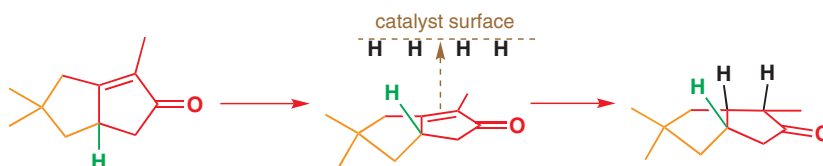
➡ You met catalytic hydrogenation in Chapter 23. For a reminder of what **stereoselective** and **stereospecific** mean, see p. 396.

Almost any *cis*-fused junction from 3,3 upwards can be made. Even bicyclo[1.1.0]butane exists, although it is not very stable. *cis*-Fused 4,5, 4,6, and 5,5 systems are common and are much more stable than their *trans* isomers.

Any method of making such bicyclic compounds will therefore automatically form this stereochemistry. Consider this hydrogenation:

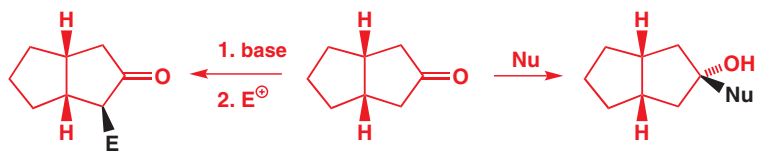


The two new hydrogen atoms (shown in black) must, of course, add *cis* to one another: this is a consequence of the stereospecificity of the reaction. What is interesting is that they have also added *cis* to the green hydrogen atom that was already there. This approach does give the more stable *cis* ring junction but the stereochemistry really arises because the other ring hinders approach to the other face of the alkene. Think of it in the way illustrated below: the alkene has two different faces. On one side there is the green hydrogen atom, and on the other the orange parts of the second ring. To get hydrogenated, the alkene must lie more or less flat on the catalyst surface and that is easier on the top face as drawn.

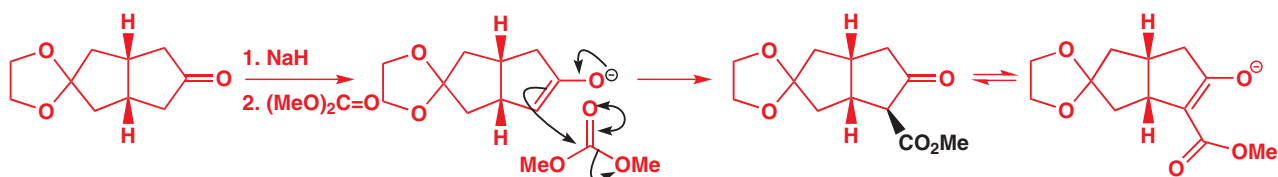


You can think of *cis*-fused rings as looking like a butterfly or an open book. The key to stereoselectivity in their reactions is that everything happens on the outside (on the cover of the book—the *exo* face). Nucleophiles add to carbonyl groups from the outside, enolates react with alkyl halides or Michael acceptors on the outside, and alkenes react with peroxyacids on the

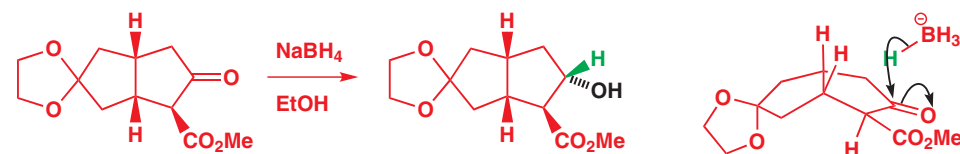
outside. Notice that this means the same side as the substituents at the ring junction. The rings are folded away from these ring-junction substituents, which are also on the outside.



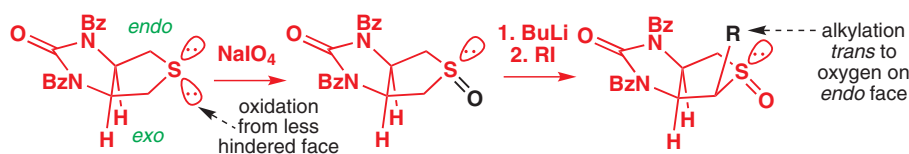
A real example comes in the acylation (Chapter 26) of the enolate from the keto-acetal below. The molecule is folded downwards and the enolate is essentially planar, so the outside face is the top face as drawn. Addition presumably occurs entirely from the outside, although the final stereochemistry of the product is controlled thermodynamically because of reversible enolization of the product, allowing the black ester group to adopt the less hindered outside position.



Reduction of the ketone product also occurs exclusively from the outside and this has the surprising effect of pushing the new OH group into the inside position. Attack from the inside is very hindered in this molecule because one of the acetal oxygen atoms is right on the flight path.

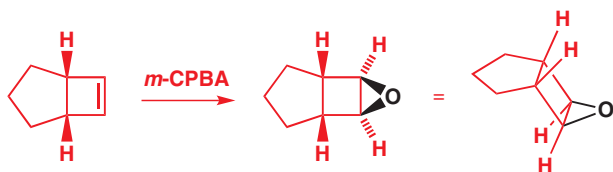


The important metabolite biotin has a *cis* bicyclic structure in which an alkyl chain lies on the more hindered face of the molecule, and any successful synthesis has to address this particular problem. You saw in Chapter 27 that sulfur stabilizes an adjacent anion, but the direct alkylation of the sulfide below is no good because the new alkyl group will go *exo*. Instead, the sulfide was oxidized to a sulfoxide from the *exo* face, giving an 8:1 ratio of *exo:endo* sulfoxides. Alkylation of the cyclic sulfoxide results in *trans* stereochemistry between the new alkyl group and the sulfoxide oxygen atom, forcing formation of the desired (*endo*) product. The synthesis is diastereoselective—but not enantioselective since there is no way of distinguishing the left and right sides of the symmetrical sulfoxide.



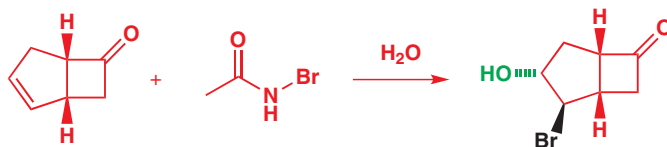
■ You will see in a moment more ways of forcing groups onto the inside face of bicyclic molecules.

A simple example of epoxidation occurs with a cyclobutene fused to a five-membered ring. This is a very rigid system and attack occurs exclusively from the outside to give a single epoxide in good yield.

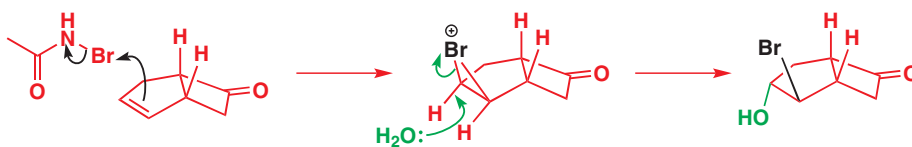


Epoxidation is stereospecific and *cis*—both new C–O bonds have to be on the same face of the old alkene. But Chapter 19 introduced you to several electrophilic additions to alkenes that were stereospecific and *trans*, many of them proceeding through a bromonium ion. If stereospecific *trans* addition occurs on a *cis*-fused bicyclic alkene, the electrophile will first add to the outside of the molecule, meaning the nucleophile will then be forced to add from the inside. A telling example occurs when the 5,4 fused unsaturated ketone below is treated with *N*-bromoacetamide in water.

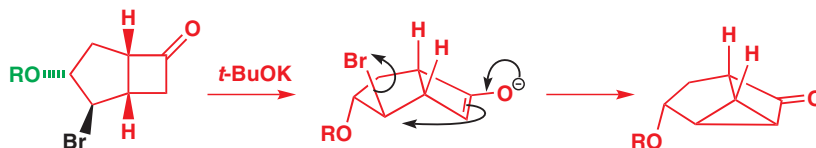
■ *N*-Bromoacetamide, like NBS (p. 836), simply provides Br₂ in low concentration.



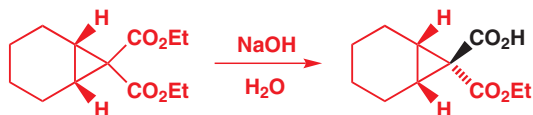
The bromonium ion is formed on the outside of the rigid structure and the water is then forced to add from the inside to get *trans* addition. As well as exhibiting stereospecificity (*trans* addition) and stereoselectivity (bromonium forms on outside), this reaction also exhibits regioselectivity in the attack of water on the bromonium ion. Water must come from inside, and it attacks the less hindered end of the bromonium ion.



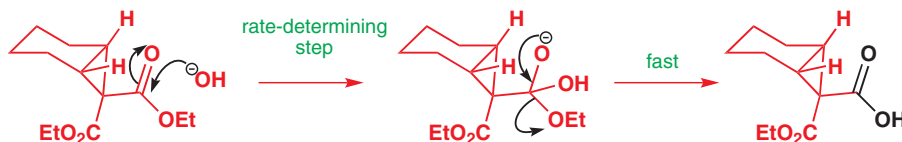
After protection of the OH group, treatment with base closes a three-membered ring to give a remarkably strained molecule. The ketone forms an enolate and the enolate attacks the alkyl bromide intramolecularly to close the third ring. This enolate is in just the right position to attack the C–Br bond from the back, precisely because of the folding of the molecule.



Inside/outside selectivity may allow the distinction between two otherwise similar functional groups. The *cis*-fused bicyclic diester below may look at first rather symmetrical but ester hydrolysis leaves one of the two esters alone while the other is converted to an acid.



Only the outside ester—on the same side as the ring junction hydrogens—is hydrolysed. In the mechanism for ester hydrolysis, the rate-determining step is the attack by the hydroxide ion so the functional group *increases* in size in the rate-determining step. This will be much easier for the ester in the outside than for the one inside the half-open book.



The end result is again that the larger of the two groups is on the inside! There are other ways to do this too. If we alkylate the enolate of a bicyclic lactone, the alkyl group (black) goes on

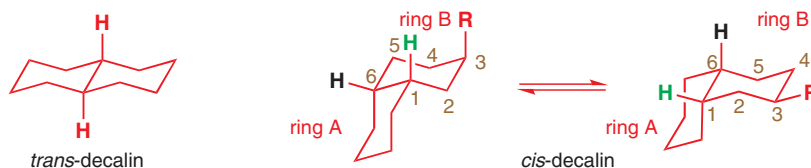
the outside as expected. But what will happen if we repeat the alkylation with a different alkyl group? The new enolate will be flat and the stereochemistry at the enolate carbon will be lost. When the new alkyl halide comes in, it will approach from the outside (green) and push the alkyl group already there into the inside.



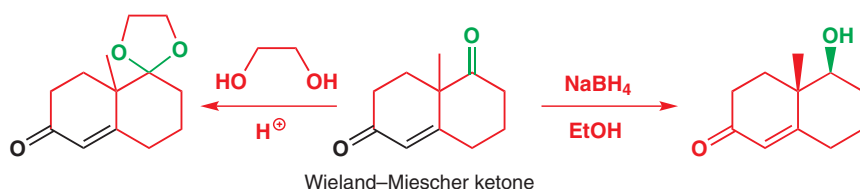
Should you wish to reverse the positions of the two groups, you simply add them in the reverse order. Whichever group is added first finishes on the inside; the other finishes on the outside.

Reactions of *cis*-decalins

You saw in Chapter 16 that while *trans*-decalins are rigid, *cis*-decalins can flip rapidly between two all-chair conformations. During the flip, all substituents change their conformation. The substituent R is axial on ring B in the first conformation of *cis*-decalin shown below but equatorial in the second. The ring junction Hs are always axial on one ring and equatorial on the other. The green hydrogen is equatorial on ring A and axial on ring B in the first conformation and vice versa in the second. Of course, they are *cis* in both. Because R gets equatorial, the second conformation is preferred in this case.

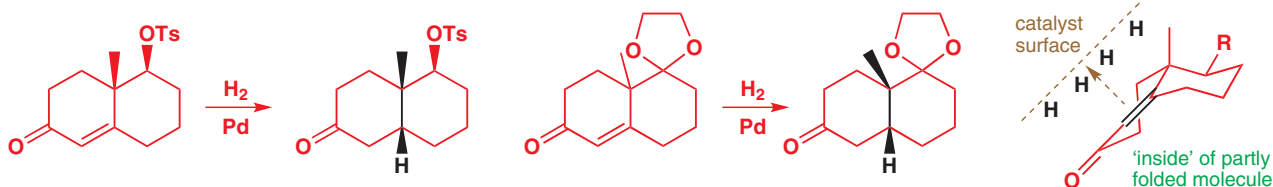


A standard reaction that gives substituted decalins is the Robinson annelation (Chapter 26). A Robinson annelation product available in quantity is the keto-enone known sometimes as the **Wieland–Miescher ketone** and used widely in steroid synthesis. The non-conjugated keto group can be protected or reduced without touching the more stable conjugated enone.

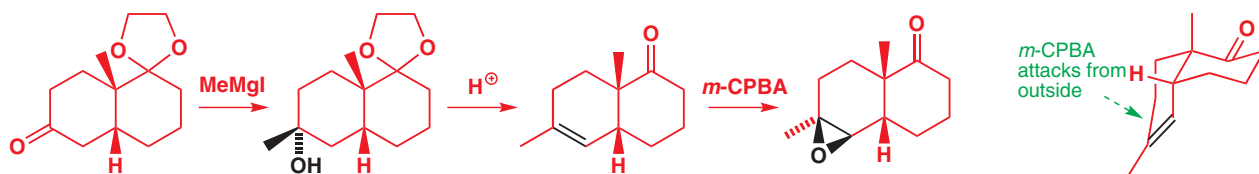


➔ The synthesis of this ketone can be found in Chapter 26, p. 638.

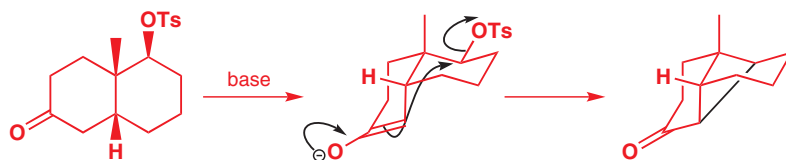
If either of these products is reduced with hydrogen and a Pd catalyst (the alcohol is first made into a tosylate), the *cis*-decalin is formed because the enone, although flattened, is already folded to some extent. A conformational drawing of either molecule shows that the top surface is better able to bind to the flat surface of the catalyst.



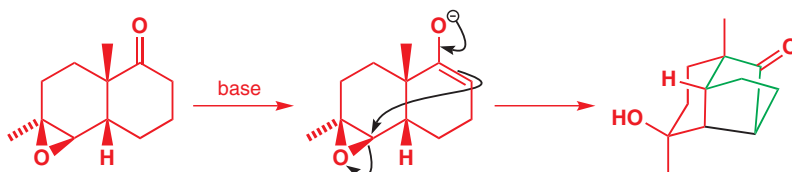
Each of these products shows interesting stereoselective reactions. The ketal can be converted into an alkene by Grignard addition and E1 elimination, and then epoxidized. Everything happens from the outside as expected, with the result that the methyl group is forced inside at the epoxidation stage.



Treatment of the other product, the keto-tosylate, with base leads to an intramolecular enolate alkylation—a cyclization on the inside of the folded molecule that actually closes a four-membered ring. The reaction is easily seen in conformational terms and the product cannot readily be drawn in conventional diagrams.



A similar reaction happens on the epoxide to produce a beautiful cage structure. This time it is a five-membered ring that is formed, but the principle is the same—the molecule closes across the fold rather easily. The new stereogenic centres can only be formed with this configuration: no other stereoisomer would be a feasible structure.



■ Notice how the ring in green has to go into a boat conformation for cyclization to be possible. This is unfavourable but still better than any intermolecular reaction.

• A summary of stereoselective reactions that occur on *cis*-fused rings

1 Reactions on the outside.

- Nucleophilic additions to carbonyl groups in the ring.
- Reactions of enolates of the same ketones with electrophiles: alkyl halides, aldols, Michael additions.
- *cis*-Additions to cyclic alkenes: hydrogenation, hydroboration, epoxidation.

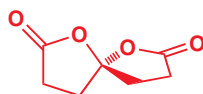
2 Reactions on the outside and then the inside.

- *trans*-Additions to cyclic alkenes: bromination, epoxidation, and epoxide opening.

3 Reactions on the inside.

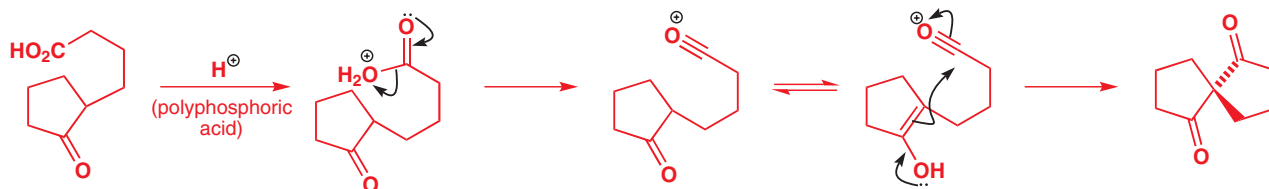
- Bond formation across the ring(s).

Spirocyclic compounds

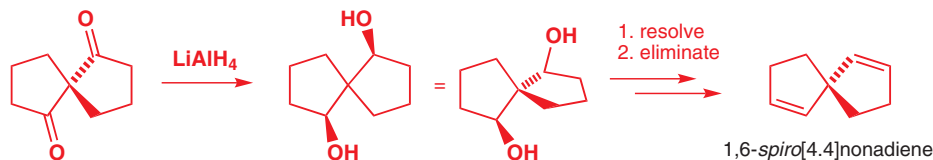


Spirocyclic rings meet at one single atom. This means that the two rings are orthogonal about the tetrahedral atom that is common to both. Even symmetrical-looking versions are unexpectedly chiral. The compound in the margin, for example, is not superimposable on its mirror image, and its symmetry is similar to that of an allene (see Chapter 14).

These sorts of compounds may look rather difficult to come by, but some simple ones are readily made. Cyclization of this keto-acid with polyphosphoric acid leads to a spirocyclic diketone. The *spiro* compound is formed because the more substituted enol is preferred in acid solution.



It is much more difficult to pass stereochemical information from one ring to the other in spirocyclic compounds because of the orthogonality of the rings. Still, some reactions are surprisingly stereoselective—one such is the reduction of the spirocyclic diketone that we made a moment ago. Treatment with LiAlH_4 gives one diastereoisomer of the spirocyclic diol.



The diol can be resolved and used to make the very simple spiro-diene as a single enantiomer. The diene is chiral even though it has no chiral centre because it does not have a plane of symmetry.

■ In Chapter 14 we explained that planes of symmetry, not chiral centres, are the things to look for when deciding whether or not a compound is chiral.

Reactions with cyclic intermediates or cyclic transition states

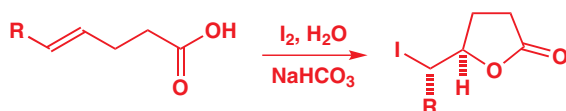
Rings are so good at controlling stereochemistry (as you have seen) that it's well worth introducing them where they are not really necessary in the final product, simply in order to enjoy those high levels of stereochemical control. In the rest of this chapter we shall consider the use of temporary rings in stereochemical control: these might be cyclic intermediates in a synthetic pathway, or cyclic reaction intermediates, or even merely cyclic transition states. All aid good stereocontrol. We shall concentrate on examples where the ring reverses the normal stereoselectivity so that some different result is possible.

Tethered functional groups can reach only one side of the molecule

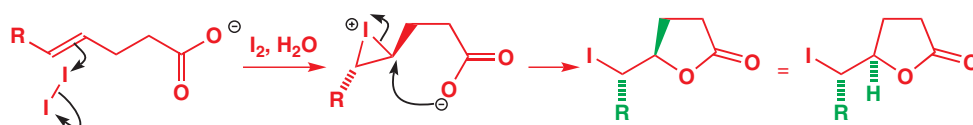
The proverbial donkey starved to death in the field with two heaps of hay because it could not decide which one to go for first. If the donkey had been tethered to a stake near one heap it would have been able to reach that heap alone and it could have feasted happily.

This principle can be applied to molecules. If a nucleophile is joined to the group it is to attack by a short chain of covalent bonds, it may be able to reach only one side. We can illustrate this idea with a reaction you met in Chapter 24: iodolactonization. To remind you, iodolactonization involves treating a non-conjugated unsaturated acid with iodine in aqueous NaHCO_3 . The product is an iodolactone.

➡ Iodolactonization is described on p. 569.



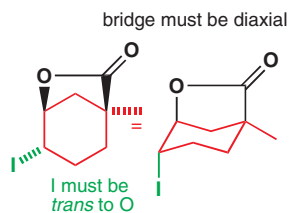
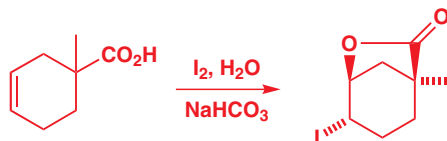
The cyclization reaction is a typical two-stage electrophilic addition to an alkene (Chapter 19) with attack by the nucleophile at the more substituted end of the intermediate iodonium ion. The ring opening is a stereospecific $\text{S}_\text{N}2$ and the stereochemistry of the alkene will be reproduced in the product.



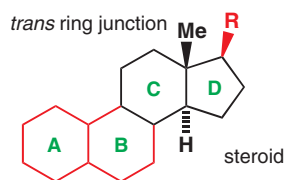
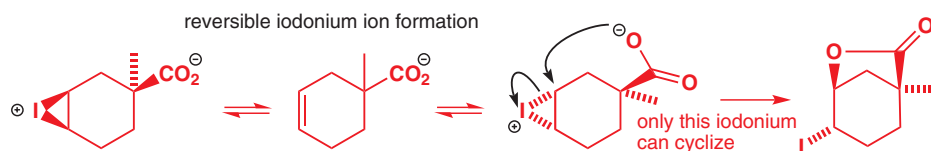
The starting acid contains an *E* alkene, giving a *trans* iodonium ion. Inversion occurs in the attack of the carboxylate anion on the iodonium ion and we have shown this by bringing the nucleophile in at 180° to the leaving group, with both bonds in the plane of the paper. A single diastereoisomer of the iodolactone results from a stereospecific reaction.

Things get more interesting again when the starting material is cyclic. The iodolactonization below gives only one diastereoisomer.

■ Chapter 34 describes how to make the unsaturated six-membered starting material.



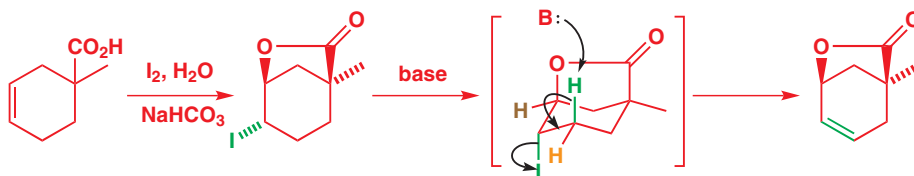
The relationship between the two stereogenic centres on the old alkene is not an issue—inversion during opening of the iodonium ion means that the I and the O must lie *trans*. But during the cyclization the carboxylic acid can attack only the nearer side of what was the double bond—in other words the bridge in black has no choice but to be *cis* across the red six-membered ring. The reason for this is that, while formation of the iodonium ion is reversible, only the iodonium ion with the I and CO₂H groups *trans* to each other can cyclize. Tethering the nucleophilic CO₂H group to the alkene dictates the stereochemistry of the product.



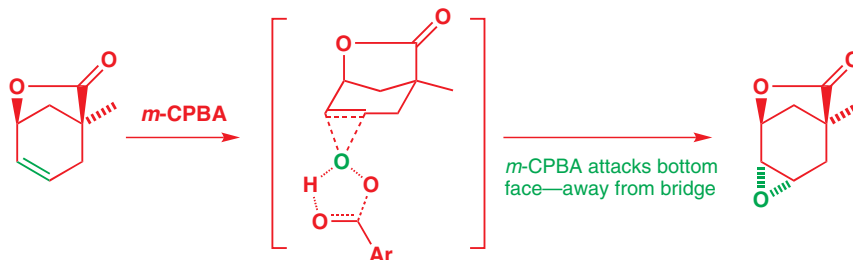
This reaction can be used to solve a general problem in the synthesis of steroids: the construction of a diketone with *trans*-fused 6,5 rings and a quaternary carbon atom at the ring junction. One solution to this problem uses the lactone just made.

The lactone makes a good temporary tether because it can be hydrolysed or reduced to break the ring at the C–O bond and reveal new stereogenic centres on the old structure. In this sequence the lactone ring controls all the subsequent stereochemistry of the molecule in two ways: it fixes the conformation rigidly in one chair form—hence forcing the iodide to be axial—and it blocks one face of the ring. From the lactone above, an alkene is introduced by E2 reaction on the iodide. This stereospecific reaction requires an anti-periplanar H atom so it has to take the only available neighbouring axial hydrogen atom, shown in green. The brown and orange hydrogens are not anti-periplanar and anyway elimination with the brown one would produce a bridgehead alkene.

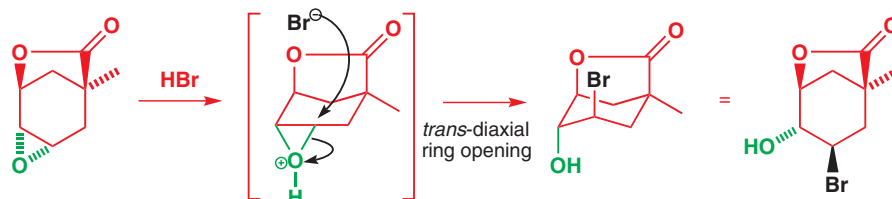
➡ The impossibility of bridgehead alkenes is mentioned in Chapter 17.



The resulting alkene has its top face blocked by the lactone bridge so epoxidation occurs entirely from the bottom face.

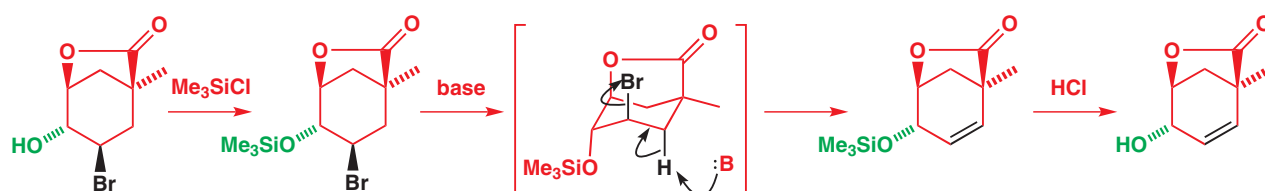


Now the epoxide is opened with HBr. Only the *trans* diaxial opening product is possible, so the bromide ion is forced to attack from the top face.

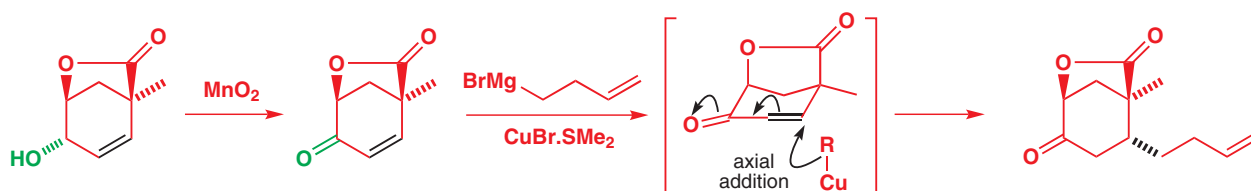


■ This is an epoxide of the type we discussed on p. 836: it has no choice but to open to the *trans* diaxial product.

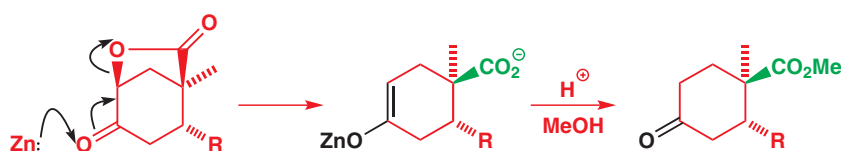
Do you see how the functional groups are being pushed round the ring? This process is extended further by a second elimination, after protection, which again seeks out the only neighbouring axial hydrogen.



The protecting silyl group is removed in acid, ready for the next important reaction: a Michael addition requiring the alcohol to be oxidized to a ketone. Allylic (or benzylic) alcohols can be oxidized by manganese dioxide, and with three atoms now trigonal the ring becomes even further flattened. But-3-enyl Grignard reagent is added with Cu(I) catalysis to make sure that conjugate addition occurs. Conjugate addition normally gives the axial product, as we saw earlier, and fortunately this is not the direction blocked by the bridge.

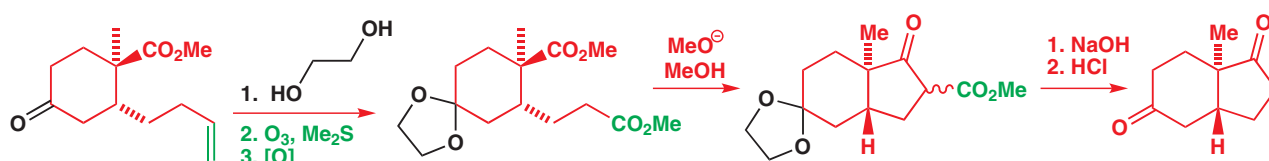


The bridge has now done its work and is removed by zinc metal reduction. This reaction removes leaving groups on the atoms next to carbonyl groups. In this case it is the axial carboxylate that is driven out by the zinc. The released carboxyl group is esterified.



■ This may look like a new reaction but think back to the Reformatsky reaction (Chapter 26, p 631). Both form zinc enolates from carbonyl compounds with adjacent leaving groups.

The last stages are shown below. The ketone is protected, and the alkene oxidized to a carbonyl group by ozonolysis (Chapter 19). The diester can be cyclized by a Claisen ester condensation (Chapter 26). The stereogenic centres in the ring are not affected by any of these reactions so a *trans* ring junction must result from this reaction. After ester hydrolysis, HCl decarboxylates the product and removes the protecting group.



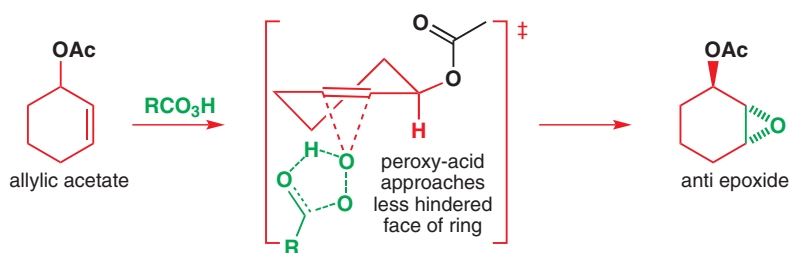
It is not easy to set up a *trans*-fused 5,6 system. In this sequence the molecule is effectively tricked into making the *trans* ring junction by the work done with the lactone tether.

Cyclic transition states can reverse normal stereoselectivity

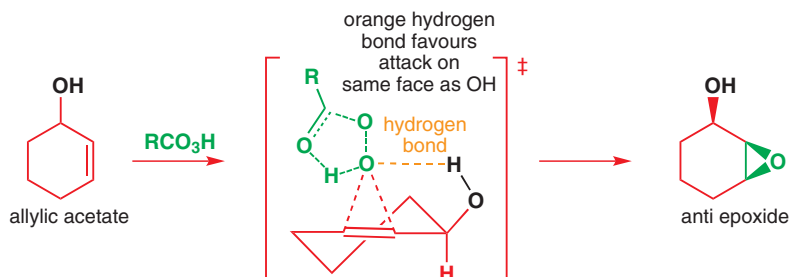
Formation of a ring in an intermediate is a means of enforcing a certain stereochemistry—the example you have seen made use of a lactone. But even transient formation of a ring in a cyclic transition state can be enough to control stereochemistry highly effectively. You will see further examples in the next chapter, but here we just present one type of reaction with this property: epoxidation.

Of course epoxidation reactions *form* rings, and you have seen examples of epoxidations with *m*-CPBA even in this chapter (p. 848) of alkenes such as cyclohexene. We pointed out in Chapter 19 that epoxidation is stereospecific because both new C–O bonds form to the same face of the alkene.

If we block one face of the ring with a substituent—even quite a small one, such as an acetate group—epoxidation becomes stereoselective for the face *anti* to the substituent already there.

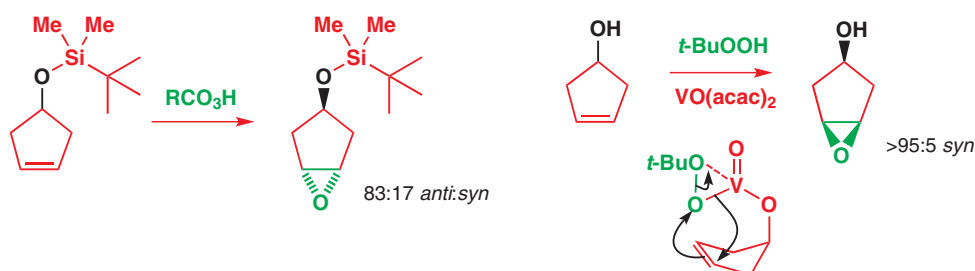


But there is one important exception to this rule, when the substituent is a hydroxyl group. When an allylic alcohol is epoxidized, the peroxy-acid attacks the face of the alkene *syn* to the hydroxyl group, even when that face is more crowded. For cyclohexenol the ratio of *syn* epoxide to *anti* epoxide is 24:1 with *m*-CPBA and it rises to 50:1 with $\text{CF}_3\text{CO}_3\text{H}$.



The reason is shown in the transition state: the OH group can hydrogen bond, through the H of the alcohol, to the peroxy-acid, stabilizing the transition state when the epoxidation is occurring *syn*. This hydrogen bond means that peroxy-acid epoxidations of alkenes with adjacent hydroxyl groups are much faster than epoxidations of simple alkenes, even when no stereochemistry is involved.

Peroxy-acids work for epoxidizing allylic alcohols *syn* to the OH group, but another reagent is better when the OH group is further from the alkene. 4-Hydroxycyclopentene, for example, can be converted into either diastereomer of the epoxide. If the alcohol is protected with a large group such as TBDMS (*t*-butyl-dimethylsilyl), it becomes a simple blocking group and the epoxide is formed on the opposite face of the alkene. The selectivity is reasonable (83:17) given that the blocking group is quite distant. But if the OH group is not blocked at all but left free, and the epoxidation reagent is the vanadium complex $\text{VO}(\text{acac})_2$ combined with *t*-BuOOH, the *syn* epoxide is formed instead. The vanadyl group chelates reagent and alcohol, and delivers the reactive oxygen atom to the same face of the alkene as shown.



In both epoxidation examples, the stereoselectivity is due to the cyclic nature of the transition state: the fact that there is a hydrogen bond or O–metal bond ‘delivering’ the reagent to one face of the alkene. Effectively we have moved on from the tethered *nucleophiles* of the last section to (transiently) tethered *reagents*. This is a very important concept, and we revisit it in the next chapter: cyclic transition states are the key to getting good stereoselectivity in reactions of acyclic compounds.

To summarize...

Diastereoselectivity in rings generally follows a few simple principles:

- Flattened three-, four-, or five-membered rings, especially ones with two or more trigonal carbons in the ring, are generally attacked from the less hindered face.
- Flattened six-membered rings with two or more trigonal carbons in the ring (that is, which are not already a chair—so six-membered rings with one trigonal C atom don’t count here) react in such a way that the product becomes an axially substituted chair.
- Bicyclic compounds react on the outside face.
- Reaction on the more hindered face can be encouraged by: (1) tethered nucleophiles or (2) cyclic transition states (tethered reagents).

Diastereoselectivity in compounds without rings is different: it is less well controlled because there are many more conformations available to the molecule. But even in acyclic compounds rings can still be important, and some of the best diastereoselectivities arise when there is a ring formed temporarily in the transition state of the reaction. With or without cyclic transition states, in some cases we have good prospects of predicting which diastereoisomer will be the major reaction product, or explaining the diastereoselectivity if we already know this. That is the subject of the next chapter.

VO(acac)₂

Vanadyl (acac)₂ is a square pyramidal complex of two molecules of the enolate of ‘acac’ (acetyl acetone, pentan-2,4-dione) and the vanadyl (V=O) dication. It can easily accept another ligand to form an octahedral complex so there is plenty of room for the alcohol to add and for the *t*-BuOOH to displace one of the ‘acac’ ligands to give some complex with the essential ingredients for the reaction here.

■ Another reaction in which an allylic alcohol is extremely effective at delivering a reagent via a cyclic transition state is the cyclopropanation reaction known as the Simmons–Smith reaction, described on p. 1017.

Further reading

Oxford Primers by A. J. Kirby, *Stereoelectronic effects*, OUP, Oxford, 1996, and M. Grossel, *Alicyclic Chemistry*, OUP, Oxford, 1997 are relevant. The most comprehensive text is E. L. Eliel and S. H. Wilen, *Stereochemistry of Organic Compounds*, Wiley Interscience, New York, 1994.

The elegant work of Jeffrey Aubé, describing the selective formation of substituted piperidines by control of their conformation, is in *Angew. Chem. Int. Ed.* 2011, **50**, 2734.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book’s Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>